

3.1.3 Enhancement of materials

Content	Potential links to maths and science
Students are expected to be able to describe enhancement methods for given materials and explain their suitability for specific product applications.	Understand the appropriate use of materials, including polymers, composites, woods and based on their physical properties.

Polymer enhancement

Content	Potential links to maths and science
The use of additives to enhance properties, including: • UV stabilisers to prolong the life of polymers • bio-batch materials to encourage biodegradability. Students should be familiar with how additives are used in specific polymer products, eg patio furniture, food packaging and carrier bags.	

Wood enhancement

Content	Potential links to maths and science
The combining of natural timber with resins and lamination to give enhanced properties, eg increased strength and stability. Enhancing timber products with preservatives, finishes and coatings.	

Metal enhancement

Content	Potential links to maths and science
Students should be aware of heat treatment methods of enhancing metals, including: • case hardening • hardening and tempering.	